

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

PRE-MEDICAL : NURTURE PHASE-I & II

TEST # 07

DATE : 09-09-2018

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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Your Target is to secure Good Rank in NEET 2020

MINOR TEST # 07	Topics	DATE : 09 - 09 - 2018
PRE-MEDICAL : NURTURE COURSE : PHASE – I & II		
PHYSICS	: NLM & Friction ,WPE	
CHEMISTRY	: Mole Concept, Atomic Structure, Periodic Table, Chemical Bonding	
BIOLOGY	: Plant Morphology (Till seed), Cockroach And Frog	

Next Up Coming Test

MINOR TEST # 08	Topics	DATE : 30 - 09 - 2018
PRE-MEDICAL : NURTURE COURSE : PHASE – I & II (PATTERN : NEET)		
PHYSICS	: Work, Power & Energy, Circular Motion	
CHEMISTRY	: Gaseous State, Chemical & Ionic Equilibrium	
BIOLOGY	: Families of Angiosperm, Anatomy of flowering Plants (up to vascular bundles), Digestion and Absorption	

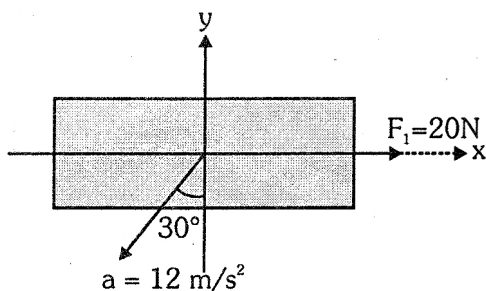
HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. Three forces act on a particle that moves with unchanging velocity $\vec{v} = (3\hat{i} - 4\hat{j})$ m/s. Two of the forces are $\vec{F}_1 = (3\hat{i} + 2\hat{j} - 4\hat{k})$ N and $\vec{F}_2 = (-5\hat{i} + 8\hat{j} - 3\hat{k})$ N. The third force is :-

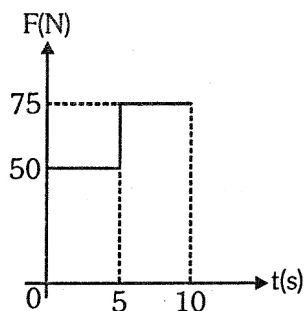
- (1) $(-2\hat{i} + 10\hat{j} - 7\hat{k})$ N
(2) $(2\hat{i} - 10\hat{j} + 7\hat{k})$ N
(3) $(7\hat{i} - 2\hat{k} + 10\hat{j})$ N
(4) none of these

2. There are two forces on the 2.0 kg box in the overhead view of figure but only one is shown. The figure also shows the acceleration of the box. The second force is nearly :-



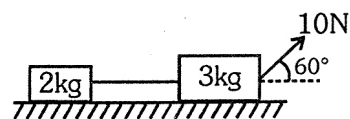
- (1) $-20\hat{j}$ N (2) $-20\hat{i}$ N + $20\hat{j}$ N
(3) $-32\hat{i} - 21\hat{j}$ N (4) $-21\hat{i} - 16\hat{j}$ N

3. A force represented as show in figure acts on a body having a mass of 16 kg. The velocity of the body at $t = 10$ s, if the body starts from rest:-



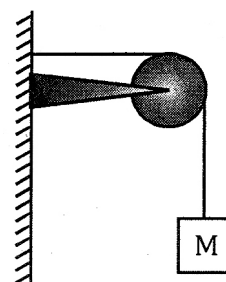
- (1) 100 m/s (2) 50 m/s
(3) 49 m/s (4) 39 m/s

4. Find the tension in the string which connected the blocks as shown in the following figure :-



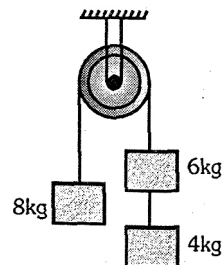
- (1) 2 N (2) 3 N
(3) 5 N (4) 10 N

5. A string of negligible mass going over a clamped pulley of mass m supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given by :-



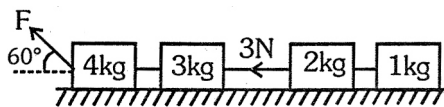
- (1) $\sqrt{2}$ Mg
(2) $\sqrt{2}$ mg
(3) $\left(\sqrt{(M+m)^2 + m^2}\right)g$
(4) $\left(\sqrt{(M+m)^2 + M^2}\right)g$

6. Three blocks of masses 4 kg, 6 kg and 8 kg are hanging over a fixed pulley as shown. The tension in the the string connecting 4 kg and 6 kg block is :- ($g = 10$ m/s²)

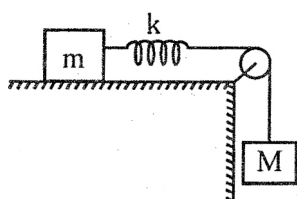


- (1) 4 N (2) 6 N
(3) $\frac{320}{9}$ N (4) $\frac{40}{9}$ N

7. Figure shows four blocks that are being pulled along a smooth horizontal surface. The masses of the blocks and tension in one cord are given. The pulling force F is :-



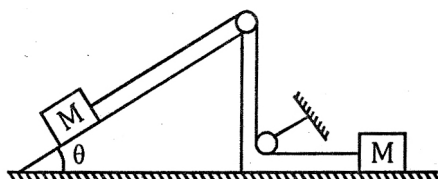
- (1) 5 N (2) 10 N
(3) 12.5 N (4) 20 N
8. Two masses, M and m are connected together by a pulley two strings and a stretched spring of force constant k as shown. Assume that string, pulley and a spring all are massless and surface below m is smooth. The amount by which the spring is stretched:-



- (1) $\frac{mMg}{k(m+M)}$ (2) $\frac{2mMg}{k(m+M)}$
(3) $\frac{Mg}{k}$ (4) $\frac{(m+M)g}{2k}$
9. A block of mass M is pulled along a horizontal smooth surface by a rope of mass m . Force P is applied at one end of the rope. The force which the rope exerts on the block is :-

- (1) $\frac{P}{(M-m)}$ (2) $\frac{PM}{(M+m)}$
(3) $\frac{Pm}{(M+m)}$ (4) $P \frac{m}{M}$

10. Two blocks, each having a mass M , rest on frictionless surface as shown in the figure. If the pulleys are light and frictionless, and M on the incline is allowed to move down, then the tension in the string will be :-



- (1) $2.3 Mg \sin \theta$ (2) $1.3 Mg \sin \theta$
(3) $\frac{Mg}{2} \sin \theta$ (4) $2 Mg \sin \theta$

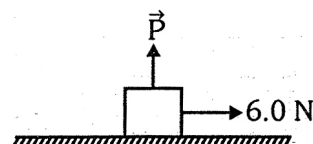
11. With what minimum acceleration can a fireman slide down a rope whose breaking strength is $3/4$ th of his weight.

- (1) $\frac{g}{4} \text{ m/s}^2$ (2) $g \text{ m/s}^2$
(3) $\frac{3}{4} \text{ m/s}^2$ (4) zero

12. A 20 kg body is pushed with just enough force to start it moving across a floor and the same force continues to act afterwards. The coefficient of static and kinetic friction are 0.6 and 0.2 respectively. The acceleration of the body is :- ($g = 10 \text{ m/s}^2$)

- (1) 1 m/s^2 (2) 2 m/s^2
(3) 3 m/s^2 (4) 4 m/s^2

13. A 2.5 kg block is initially at rest on a horizontal surface. A 6.0 N horizontal force and a vertical force $\vec{P}$ are applied to the block as shown in figure. The coefficient of friction for the block and surface is 0.4. The magnitude of friction force when $P = 9 \text{ N}$ ($g = 10 \text{ m/s}^2$)

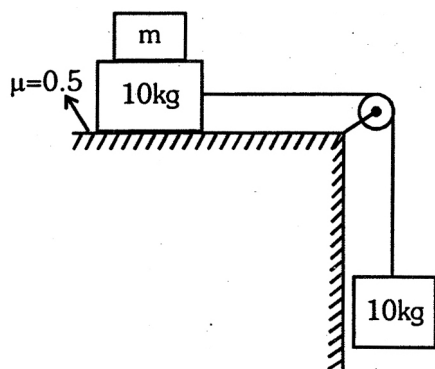


- (1) 6.0 N (2) 6.4 N
(3) 9.0 N (4) zero

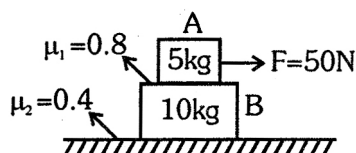
14. Smooth Find maximum force for which both the block will move with same acceleration.

- (1) 12 N (2) 24 N
(3) 36 N (4) 42 N

15. Find the minimum value of m that should be placed on 10kg . So that system remain at rest.



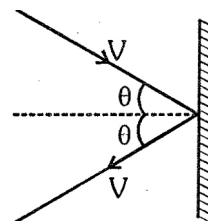
- (1) 5 kg (2) 10 kg
(3) 15 kg (4) 20 kg
16. Find acceleration of block A and B. If $\mu_1 = 0.8$ and $\mu_2 = 0.4$



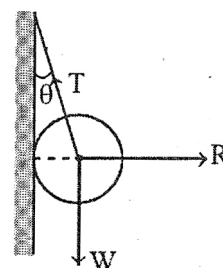
- (1) 0 m/s^2 and 0 m/s^2
(2) 1 m/s^2 and $1/2 \text{ m/s}^2$
(3) 2 m/s^2 and 0 m/s^2
(4) $1/2 \text{ m/s}^2$ and $1/4 \text{ m/s}^2$
17. When a horse pulls a wagon, the force that causes the horse to move forward is the force
- (1) He exerts on the wagon
(2) The wagon exerts on him
(3) The ground exerts on him
(4) He exerts on the ground
18. If the force of gravity suddenly disappears :-
- (1) The mass of all bodies will become zero
(2) The weight of all bodies will become zero
(3) Both mass and weight of all bodies will become zero
(4) Neither mass nor weight of all bodies will become zero
19. The velocity acquired by a mass m in travelling a certain distance d starting from rest under the action of a constant force is directly proportional to :-

- (1) $\sqrt{m}$ (2) m°
(3) $\frac{1}{\sqrt{m}}$ (4) m

20. A water jet, whose cross sectional area is ' a ' θ ' with the normal and rebounds elastically. The velocity of water of density ' d ' is v . Force exerted on wall is :-



- (1) $2 av^2d \cos\theta$ (2) $2 av^2d \sin\theta$
(3) $2 avd \cos\theta$ (4) $avd \cos\theta$
21. A player catches a ball of 200 g moving with a speed of 20 m/s . If the time taken to complete the catch is 0.5 s , the force exerted on the player's hand is :-
- (1) 8 N (2) 4 N (3) 2 N (4) 0
22. If force $F = 500 - 100t$, then impulse as a function of time will be :-
- (1) $500t - 50t^2$ (2) $50t - 10$
(3) $50 - t^2$ (4) $100t^2$
23. A metal sphere is hung by a string fixed to a wall. The forces acting on the sphere are shown in fig. Which of the following statements is correct ?

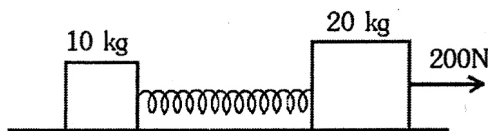


- (a) $\vec{R} + \vec{T} + \vec{W} = 0$ (b) $T^2 = R^2 + W^2$
(c) $T = R + W$ (d) $R = W \tan \theta$
(1) a, b, c (2) b, c, d
(3) a, b, d (4) a, b, c, d

24. If a parrot starts flying upwards with an acceleration in an air tight cage, then the boy will feel the weight of the cage:

(1) Unchanged
(2) Reduced
(3) Increased
(4) Nothing can be said

25. Two masses of 10 kg and 20 kg respectively are connected by a massless spring as shown in the figure. A force of 200 N acts on the 20 kg mass. At the instant shown the 10 kg mass has an acceleration 4 m/s^2 rightwards. What is the acceleration of 20 kg mass ?

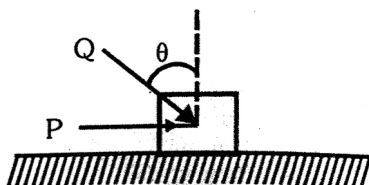


(1) Zero (2) 10 m/s^2
(3) 4 m/s^2 (4) 8 m/s^2

26. A block slides with constant velocity on a plane inclined at an angle θ . The same block is pushed up the plane with an initial velocity v_0 . The distance covered by the block before coming to rest is :-

(1) $\frac{v_0^2}{2g \sin \theta}$ (2) $\frac{v_0^2}{4g \sin \theta}$
(3) $\frac{v_0^2 \sin^2 \theta}{2g}$ (4) $\frac{v_0^2 \sin^2 \theta}{4g}$

27. A block of mass m lying on a rough horizontal plane is acted upon by a horizontal force P and another force Q inclined at an angle θ to the vertical. The block will remain in equilibrium if the coefficient of friction between it and the surface is :-



(1) $\frac{P + Q \sin \theta}{mg + Q \cos \theta}$

(2) $\frac{P \cos \theta + Q}{mg - Q \sin \theta}$

(3) $\frac{P + Q \cos \theta}{mg + Q \sin \theta}$

(4) $\frac{P \sin \theta + Q}{mg - Q \cos \theta}$

28. A given object takes n times as much time to slide down a 45° rough incline as it takes to slide down a perfectly smooth 45° incline. The coefficient of kinetic friction between the object and the incline is given by :

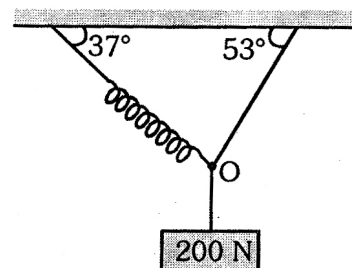
(1) $\left(1 - \frac{1}{n^2}\right)$

(2) $\left(\frac{1}{1 - n^2}\right)$

(3) $\sqrt{\left(1 - \frac{1}{n^2}\right)}$

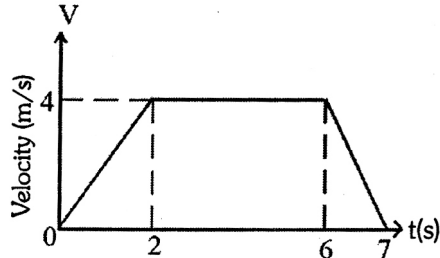
(4) $\sqrt{\left(\frac{1}{1 - n^2}\right)}$

29. In the set-up shown, a 200 N block is supported in equilibrium with the help of strings and a spring, all knotted at point O. Extension in the spring is 4 cm. Force constant of the spring is closest to [$g = 10 \text{ m/s}^2$]

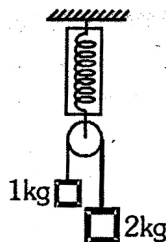


(1) 30 N/m
(2) 2500 N/m
(3) 3000 N/m
(4) 4000 N/m

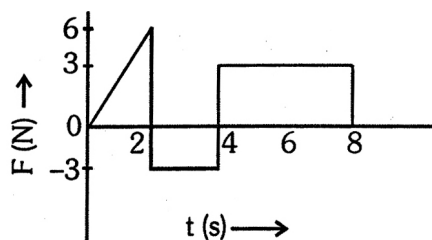
30. A lift of mass 100 kg starts moving from rests in the upward direction. Fig shows the variation if speed of the lift, T_1 , T_2 and T_3 stand for tension in the rope from zero to two seconds, two to six seconds, six to seven seconds respectively then :-



- (1) $T_1 : T_2 : T_3 :: 1 : 1 : 1$
 (2) $T_1 : T_2 : T_3 :: 6 : 5 : 3$
 (3) $T_1 : T_2 : T_3 :: 3 : 5 : 6$
 (4) $T_1 : T_2 : T_3 :: 6 : 5 : 6$
31. Two masses of 1 kg and 2 kg are attached to the ends of a massless string passing over a pulley of negligible weight. The pulley itself is attached to a light spring balance as shown in figure. The masses start moving; during this interval the reading of spring balance will be:-



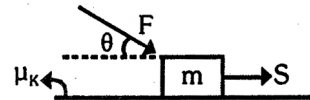
- (1) More than 3 kg.
 (2) Less than 3 kg.
 (3) Equal to 3 kg.
 (4) None of the above
32. The force 'F' acting on a particle of mass 'm' is indicated by the force-time graph shown below. The change in momentum of the particle over the time interval from zero to 8 s is :-



- (1) 24 Ns (2) 20 Ns
 (3) 12 Ns (4) 6 Ns
33. A body of mass m is displaced from point A(3, 1, 2) to point B(4, 2, 1) under the effect of a force $F = (\hat{i} + 2\hat{j} + 3\hat{k})N$, calculate W.D. by the force.

- (1) 57 J (2) 11 J
 (3) 0 (4) 22 J

34. Find work done by friction for displacement 'S' ?



- (1) $\mu_k(mg + F\sin\theta).S$
 (2) $-\mu_k(mg + F\sin\theta).S$
 (3) $\mu_k(mg - F\sin\theta).S$
 (4) $-\mu_k(mg - F\sin\theta).S$

35. A force $F = Kx^2$ acts on a particle at an angle of 60° with the x-axis. the work done in displacing the particle from x_1 to x_2 will be -

- (1) $\frac{kx^2}{2}$
 (2) $\frac{k}{2}(x_2^2 - x_1^2)$
 (3) $\frac{k}{6}(x_2^3 - x_1^3)$
 (4) $\frac{k}{3}(x_2^3 - x_1^3)$

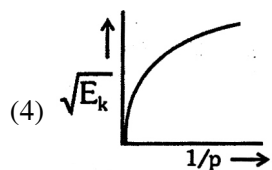
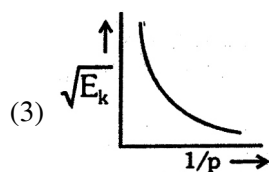
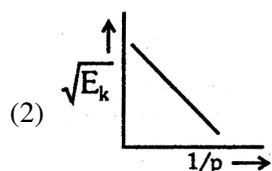
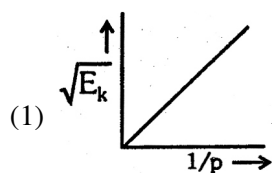
36. A force acts on a 30 g particle in such a way that the position of the particle as a function of time is given by $x = 3t - 4t^2 + t^3$, where x is in metres and t is in seconds. The work done during the first 4 second is :-

- (1) 5.28 J (2) 450 mJ
 (3) 490 mJ (4) 530 mJ

37. If K.E. increases by 4%. Then momentum will increase by :-

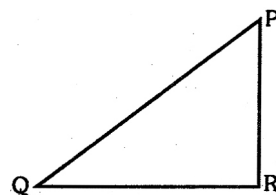
- (1) 1.5% (2) 9%
 (3) 3% (4) 2%

38. The graph between $\sqrt{E_k}$ and $\frac{1}{p}$ is
(E_k = kinetic energy and p - momentum) -



39. A 0.5 kg ball is thrown up with an initial speed 14 m/s and reaches a maximum height of 8.0 m. How much energy is dissipated by air drag acting on the ball during the ascent ?
(1) 19.6 joules
(2) 4.9 joules
(3) 10 joules
(4) 9.8 joules
40. A ball of mass 4 kg and another of mass 8 kg are dropped together from a 100 feet tall building. After a fall of 50 feet each towards earth, their respective kinetic energies will be in the ratio of :-
(1) 1 : 4
(2) 1 : 2
(3) 1 : $\sqrt{2}$
(4) $\sqrt{2}$: 1

41. For the path PQR in a conservative force field the amounts work done in carrying a body from P to Q and from Q to R are 8 Joule and 2 Joule respectively. The work done in carrying the body from P to R will be -



- (1) 10 J
(2) 6 J
(3) $\sqrt{68}$ J
(4) Zero
42. If the momentum of a body is increased n times, its kinetic energy increases.
(1) n times
(2) $2n$ times
(3) $\sqrt{n}$ times
(4) n^2 times
43. If K.E. body is increased by 100%. Then % change in 'P'.
(1) 50% (2) 41.4%
(3) 10% (4) 20%
44. A force $\vec{F} = (3x^2 + 2x - 7)$ N acts on a 2 kg body as a result of which the body gets displaced from $x = 0$ to $x = 5$ m. The work done by the force will be-
(1) 35 J (2) 70 J
(3) 115 J (4) 270 J
45. A body of mass 6 kg under a force which causes displacement in it given ' $S = \frac{t^2}{4}$ ', metres where ' t ' is time. The work done by the force in 2 seconds is :-
(1) 12 J (2) 9 J
(3) 6 J (4) 3 J

- 46.** A sample of CaCO_3 has Ca = 40%, C = 12% and O = 48%. If the law of constant proportions is true, then the mass of Ca in 5 g of CaCO_3 from another source will be:
 (1) 2.0 g (2) 0.2 g
 (3) 0.02 g (4) 20.0 g
- 47.** Number of atoms in 4.25 g of NH_3 is:
 (1) 6.023×10^{23}
 (2) $4 \times 6.023 \times 10^{23}$
 (3) 1.7×10^{24}
 (4) $4.5 \times 6.023 \times 10^{23}$
- 48.** If 6 litre of H_2 and 5.6 Litre of Cl_2 are mixed and exploded in an eudiometer, the volume of HCl formed is:
 (1) 6.0 litre (2) 5.6 litre
 (3) 11.2 litre (4) 11.6 litre
- 49.** A_1 g of an element gives A_2 g of its chloride; the equivalent mass of the element is:
 (1) $\frac{A_1}{A_2 - A_1} \times 35.5$ (2) $\frac{A_2}{A_2 - A_1} \times 35.5$
 (3) $\frac{A_2 - A_1}{A_1} \times 35.5$ (4) $\frac{A_2 - A_1}{A_2} \times 35.5$
- 50.** The largest number of molecules is in:
 (1) 28 g of CO (2) 46 g of $\text{C}_2\text{H}_5\text{OH}$
 (3) 36 g of H_2O (4) 54 g of N_2O_5
- 51.** 4.6×10^{22} atoms of an element weigh 13.8 g. The atomic mass of the element is:
 (1) 290 (2) 180
 (3) 34.4 (4) 10.4
- 52.** A gaseous mixture contains oxygen and nitrogen in the ratio of 1 : 4 by mass. Therefore, the ratio of their number of molecules is:
 (1) 1 : 4 (2) 1 : 8
 (3) 7 : 32 (4) 3 : 16
- 53.** 0.5 M H_2SO_4 is diluted from 1 litre to 10 litre, normality of the resulting solution is:
 (1) 1 N (2) 0.1 N
 (3) 10 N (4) 11 N
- 54.** The molarity of a solution containing 50 g of NaCl in 500 g of a solution and having a density of 0.936 g/cm^3 is:
 (1) 1.5M (2) 1.6 M
 (3) 1.8 M (4) 1.2 M
- 55.** How many moles and how many grams of NaCl are present in 250 mL of a 0.5 M NaCl solution?
 (1) 0.125 mol; 7.32 g
 (2) 7.32 mol; 0.125 g
 (3) 0.125 mol; 0.125 g
 (4) 7.32 mol; 7.32 g
- 56.** Molar solution means 1 mole of solute present in:
 (1) 1000 g of solvent
 (2) 1 litre of solvent
 (3) 1 litre of solution
 (4) 1000 g of solution
- 57.** 20 mL of 0.5 M HCl is mixed with 30 mL of 0.3 M HCl, the molarity of the resulting solution is:
 (1) 0.8 M (2) 0.53 M
 (3) 0.38 M (4) 0.83 M
- 58.** The frequency of the radiation having wave number 10 m^{-1} is:
 (1) 10 s^{-1} (2) $3 \times 10^7 \text{ s}^{-1}$
 (3) $3 \times 10^{11} \text{ s}^{-1}$ (4) $3 \times 10^9 \text{ s}^{-1}$
- 59.** The energy of a photon of radiation having wavelength 300 nm is:
 (1) $6.63 \times 10^{-29} \text{ J}$
 (2) $6.63 \times 10^{-19} \text{ J}$
 (3) $6.63 \times 10^{-28} \text{ J}$
 (4) $6.63 \times 10^{-17} \text{ J}$
- 60.** The maximum kinetic energy of the photoelectrons is found to be $6.63 \times 10^{-19} \text{ J}$, when the metal is irradiated with a radiation of frequency $2 \times 10^{15} \text{ Hz}$. The threshold frequency of the metal is about:
 (1) $1 \times 10^{15} \text{ s}^{-1}$
 (2) $2 \times 10^{15} \text{ s}^{-1}$
 (3) $3 \times 10^{15} \text{ s}^{-1}$
 (4) $1.5 \times 10^{15} \text{ s}^{-1}$

- 61.** The number of photons of light having wavelength 100 nm which can provide 1 J energy is nearly.
(1) 10^7 photons
(2) 5×10^7 photons
(3) 5×10^{17} photons
(4) 5×10^7 photons
- 62.** The atomic transition gives rise to the radiation of frequency (10^4 MHz). The change in energy per mole of atoms taking place would be :-
(1) 3.99×10^{-6} J
(2) 3.99 J
(3) 6.62×10^{-24} J
(4) 6.62×10^{-30} J
- 63.** The number of spectral lines that are possible when electrons in 7th shell in different hydrogen atoms return to the 2nd shell is :-
(1) 12 (2) 15 (3) 14 (4) 10
- 64.** The ratio of radii of first orbits of H, He⁺ and Li²⁺ is :-
(1) 1 : 2 : 3 (2) 6 : 3 : 2
(3) 1 : 4 : 9 (4) 9 : 4 : 1
- 65.** What is the energy in eV required to excite the electron from $n = 1$ to $n = 2$ state in hydrogen atom? (n = principal quantum number)
(1) 13.6 (2) 3.4
(3) 17 (4) 10.2
- 66.** If the kinetic energy of an electron is increased 4 times, the wavelength of the de-Broglie wave associated with it would become :-
(1) 4 times (2) 2 times
(3) $\frac{1}{2}$ times (4) $\frac{1}{4}$ times
- 67.** Velocity (u) of de-Broglie wave is given by :
(Where ν = frequency)
(1) $\frac{c^2}{\nu}$ (2) $\frac{h\nu}{mc}$
(3) $\frac{mc^2}{h}$ (4) $\nu\lambda$
- 68.** Which of the following sets of quantum number is correct for an electron in 4f-orbital?
(1) $n = 4, l = 3, m = +4, s = +1/2$
(2) $n = 4, l = 4, m = -4, s = -1/2$
(3) $n = 4, l = 3, m = +1, s = +1/2$
(4) $n = 3, l = 2, m = -2, s = +1/2$
- 69.** In the long form of periodic table, elements are arranged according to :
(1) increasing atomic number
(2) increasing atomic mass
(3) increasing metallic nature
(4) increasing electropositive nature
- 70.** The correct statements are :
(i) There are four d-block series.
(ii) Total d-block elements are 40.
(iii) First d-block series starts with element scandium and ends with zinc.
(iv) All members of d-block series are stable elements and found in nature.
(1) (i), (ii) and (iii) (2) (i) and (iii)
(3) (i) and (iv) (4) all of these
- 71.** The order of screening effect of electrons of s, p, d and f orbitals of a given shell of an atom on its outer shell electrons is :
(1) $s > p > d > f$ (2) $f > d > p > s$
(3) $p < d < s > f$ (4) $f > p > s > d$
- 72.** The atomic radius of elements of the following series should be nearly the same :
(1) Na, K, Rb, Cs (2) Na, Mg, Al, Si
(3) Fe, Co, Ni, Cu (4) F, Cl, Br, I
- 73.** The radii of F, F⁻, O and O²⁻ are in the order :
(1) $O^{2-} > F^- > F > O$ (2) $F^- > O^{2-} > F > O$
(3) $O^{2-} > O > F^- > F$ (4) $O^{2-} > F^- > O > F$
- 74.** The ionic radius of Cr is minimum in which of the following compounds ?
(1) CrF₃ (2) CrCl₃
(3) Cr₂O₃ (4) K₂CrO₄
- 75.** In which pair, the first atom or ion is not larger in size than the second?
(1) Br⁻, Br (2) Pb²⁺, Pb⁴⁺
(3) N, P (4) N, F

76. The electronic configurations for some neutral atoms are given below :
- (A) $1s^2, 2s^2 2p^6, 3s^2$ (B) $1s^2, 2s^2 2p^6 3s^1$
(C) $1s^2, 2s^2 2p^6, 3s^2 3p^2$ (D) $1s^2, 2s^2 2p^6, 3s^2 3p^3$
- Which of these is expected to have the highest second ionisation enthalpy?
- (1) (A) (2) (B) (3) (C) (4) (D)
77. Match the List-I with List-II and select the correct answer :
- | List-I | List-II |
|--|-------------|
| A. Most electronegative element | 1. Chlorine |
| B. The element having highest electron gain enthalpy | 2. Hydrogen |
| C. Most abundant element in the universe | 3. Nitrogen |
| D. Most abundant gas in atmosphere | 4. Fluorine |
- | A | B | C | D |
|-------|---|---|---|
| (1) 1 | 2 | 3 | 4 |
| (2) 4 | 3 | 2 | 1 |
| (3) 4 | 1 | 2 | 3 |
| (4) 2 | 3 | 4 | 1 |
78. In which solvent NaCl has maximum solubility?
- (1) H_2O (2) C_2H_5OH
(3) CH_3COCH_3 (4) $C_2H_5OC_2H_5$
79. Lattice energy of $BeCO_3$ (I), $MgCO_3$ (II) and $CaCO_3$ (III) is in order :-
- (1) $I < II < III$ (2) $I > II > III$
(3) $I < III < II$ (4) $II < I < III$
80. The values of electronegativity of atoms A and B are 1.20 and 4.0 respectively. The percentage ionic character of A—B bond is :
- (1) 50.0% (2) 72.2%
(3) 55.3% (4) 43.0%
81. Variable valency is shown by :-
- (1) s-block elements
(2) d-block elements
(3) p- and d-block elements
(4) all elements

82. In which of the species, the bond is non-directional?
- (1) NH_3 (2) H_2O (3) BF_3 (4) $NaCl$
83. In a triple bond, there is sharing of :-
- (1) 3 electrons (2) 4 electrons
(3) 6 electrons (4) several electrons
84. Which of the following sets of species does not follow octet rule?
- (1) $CO, PCl_5, PCl_3, AlCl_3$
(2) CO, B_2H_6, NH_3, H_2O
(3) $AlCl_3, BF_3, PCl_5, SF_6$
(4) $H_2O, NH_3, CO_2, AlCl_3$
85. The number and types of bonds present in NH_4^+ ion are :-
- (1) Four covalent bonds
(2) Four ionic bonds
(3) Three covalent and one coordinate bond
(4) Three covalent and one ionic bond
86. The electronegativity of O, F, N, Cl and H are 3.5, 4.0, 3.2, 3.0 and 2.1, respectively. The strongest bond will be :-
- (1) O—F (2) O—Cl (3) N—H (4) O—H
87. Which has maximum covalent character?
- (1) $NaCl$ (2) $SiCl_4$
(3) $AlCl_3$ (4) $MgCl_2$
88. Non-zero dipole moment is shown by :-
- (1) CCl_4 (2) CO_2

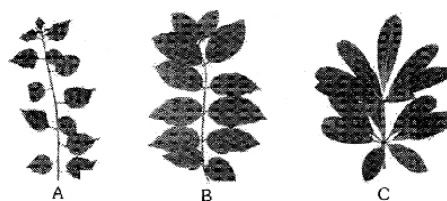


89. Which of the following are arranged in the decreasing order of dipole moment?
- (1) $CH_3I, CH_3Br, CH_3Cl, CH_3F$
(2) $CH_3Cl, CH_3F, CH_3Br, CH_3I$
(3) $CH_3Br, CH_3Cl, CH_3F, CH_3I$
(4) $CH_3Br, CH_3I, CH_3F, CH_3Cl$
90. NH_3 and BF_3 form an adduct readily because they form :-
- (1) an ionic bond
(2) a covalent bond
(3) a hydrogen bond
(4) a coordinate bond

- 91.** Regions of root from base to root tip are
 (1) Maturation zone — Cell division zone — Elongation zone
 (2) Maturation zone — Elongation zone — Cell division zone
 (3) Cell division zone — Elongation zone — Maturation zone
 (4) Elongation zone — Cell division zone — Maturation zone
- 92.** Epiphytes like *Vanda* develop special layer of absorptive tissue velamen consisting of 4 or 5 layers of long polygonal cells. Velamen is formed by
 (1) Absorbing roots
 (2) Stem
 (3) Clinging roots
 (4) Hanging roots
- 93.** Find out correct order of vegetative propagules of plants like potato, ginger, Agave, Bryophyllum and water hyacinth
 (1) Offset, bulbil, leaf bud, rhizome and eyes
 (2) Leaf bud, bulbil, offset, rhizome and eyes
 (3) Eyes, rhizome, bulbil, leaf bud and offset
 (4) Rhizome, bulbil, leaf bud, eyes and offset
- 94.** In *Opuntia*, the function of photosynthesis is carried out by
 (1) Cladode (2) Phylloclade
 (3) Phyllode (4) Bulb
- 95.** Match the following and select the correct combination from the options given below

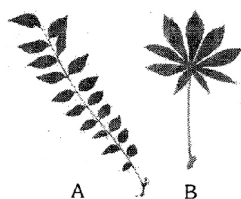
Column I (Stem Modifications)	Column II (Found in)
A. Underground stem	1. Euphorbia
B. Stem tendrils	2. Opuntia
C. Stem thorns	3. Potato
D. Flattened stem	4. Citrus
E. Fleshy cylindrical stem	5. Cucumber
(1) A-1, B-2, C-3, D-5, E-4	
(2) A-2, B-3, C-4, D-5, E-1	
(3) A-3, B-4, C-5, D-1, E-2	
(4) A-3, B-5, C-4, D-2, E-1	

- 96.** Ginger is a stem which can be differentiated from root because it
 (1) Grows parallel to ground
 (2) Stores food
 (3) Lacks chlorophyll
 (4) Has nodes and internodes
- 97.** Find the correct match
 (1) Mustard plant: leaves are opposite
 (2) Mustard plant: leaves are alternate
 (3) Guava plant: Leaves are alternate
 (4) Guava plant: Leaves are whorled
- 98.** Identify the correct types of phyllotaxy which shown in the following figures



- (1) A - Whorled, B - Alternate, C - Opposite
 (2) A - Alternate, B - Whorled, C - Opposite
 (3) A - Whorled, B - Opposite, C - Alternate
 (4) A - Alternate, B - Opposite, C - Whorled
- 99.** Study the following statements and select the correct option
 (A) Buds are present in the axil of leaflets of the compound leaf
 (B) Pulvinus leaf-base is present in some leguminous plants
 (C) In *Alstonia*, the petioles expand, become green and synthesize food
 (D) Opposite phyllotaxy is seen in guava
 (1) (B) and (D) are correct but (A) and (C) are wrong
 (2) (A) and (C) are correct but (B) and (D) are wrong
 (3) (A) and (D) are correct but (B) and (C) are wrong
 (4) (B), (C) and (D) are correct but (A) is wrong
- 100.** In sweet pea, the tendrils are modified
 (1) Stem branches (2) Leaflets
 (3) Leaves (4) Stipules

101.



See the following figures and identify leaves A and B

- (1) A - Palmately compound leaf B - Palmately compound leaf
- (2) A - Pinnately compound leaf, B - Pinnately compound leaf
- (3) A - Palmately compound leaf, B - Pinnately compound leaf
- (4) A - Pinnately compound leaf, B - Palmately compound leaf

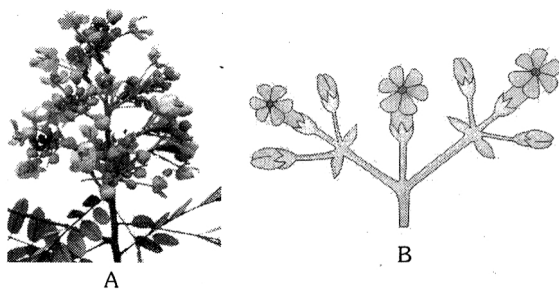
102. In cyathium inflorescence

- (1) Single male flower is surrounded by female flowers
- (2) Male and female flowers are borne in different plants
- (3) There is of one male and one female flowers
- (4) Single female flower surrounded by many peripheral male flowers

103. The special type of inflorescence in which the female flower are at bottom and male flower near ostiole and enclosed within a cup shaped fleshy thalamus (receptacle) with ostiole is called

- (1) Cyathium
- (2) Verticillaster
- (3) Spadix
- (4) Hypanthodium

104. See the following diagrams and identify inflorescence A and B



- (1) A - Cymose, B - Cymose
- (2) A - Racemose, B - Racemose
- (3) A - Racemose, B - Cymose
- (4) A - Cymose, B - Racemose

105. Butterfly shaped flower with one standard, two wing-like and two keeled petal belong to

- (1) Compositae
- (2) Rubiaceae
- (3) Malvaceae
- (4) Papilionaceae

106. In china rose the flowers are

- (1) Zygomorphic, epigynous with twisted aestivation
- (2) Actinomorphic, hypogynous with twisted aestivation
- (3) Actinomorphic, epigynous with valvate aestivation
- (4) Zygomorphic, hypogynous with imbricate aestivation

107. Glumes are modified

- (1) Petals
- (2) Bracts
- (3) Gynoecium
- (4) Androecium

108. When placenta forms a ridge along the ventral suture of the ovary and the ovules are borne on this ridge forming two rows, the placentation is termed as

- (1) Axile
- (2) Parietal
- (3) Marginal
- (4) Basal

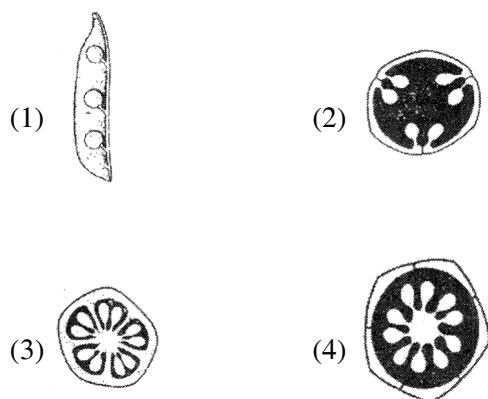
109. A flower is zygomorphic when

- (1) Any transverse section divides it into two equal halves
- (2) Only one transverse section divides it into two equal halves
- (3) Every vertical section passing through its centre divides it into two equal halves
- (4) Only one vertical section passing through its centre divides it into two equal halves

110. When the other floral parts are arranged at the base of the gynoecium, the flower is called

- (1) Hypogynous flower
- (2) Perigynous flower
- (3) Epigynous flower
- (4) Agynous flower

111. Which one of the following diagrams represents the placentation in *Dianthus*



112. When the margins of sepals or petals overlap one another without any particular direction, the condition is termed as

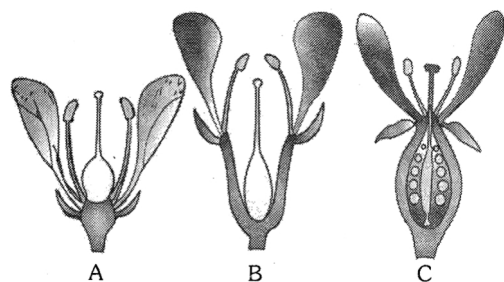
- (1) Twisted
- (2) Valvate
- (3) Vexillary
- (4) Imbricate

113. The megasporangium of the angiosperms on maturation gives rise to

- (1) A fruit
- (2) Seed
- (3) An embryo
- (4) Cotyledons

114. Match the following figures with I, II and III

- (I) Hypogynous flower
- (II) Perigynous flower
- (III) Epigynous flower



(1) A-III, B-I, C-II

(2) A-III, B-II, C-I

(3) A-I, B-III, C-II

(4) A-I, B-II, C-III

115. A drupe develops in

- (1) Tomato
- (2) Mango
- (3) Wheat
- (4) Pea

116. Figure - I - Mango, Figure - II - Coconut are shown in the following diagram. Identify the parts of the fruit A, B, C and D respectively

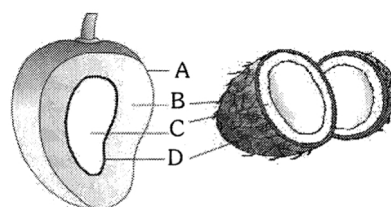


Figure - I

Figure - II

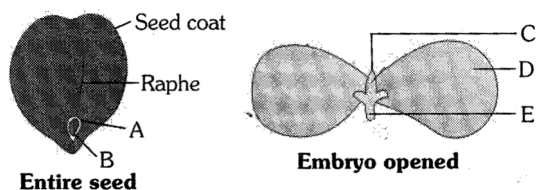
(1) Epicarp, Mesocarp, Embryo, Endocarp

(2) Epicarp, Mesocarp, Ovary, Endocarp

(3) Epicarp, Mesocarp, Ovule, Endocarp

(4) Epicarp, Mesocarp, Seed, Endocarp

117. The following diagram is the typical structure of dicotyledonous seeds. In which one of the options all the five parts A to E are correctly identified



(1) A - Hilum, B - Micropyle, C - Plumule, D - Radicle, E - Cotyledon

(2) A - Micropyle, B - Hilum, C - Plumule, D - Cotyledon, E - Radicle

(3) A - Hilum, B - Micropyle, C - Plumule, D - Cotyledon, E - Radicle

(4) A - Hilum, B - Micropyle, C - Radicle, D - Cotyledon, E - Plumule

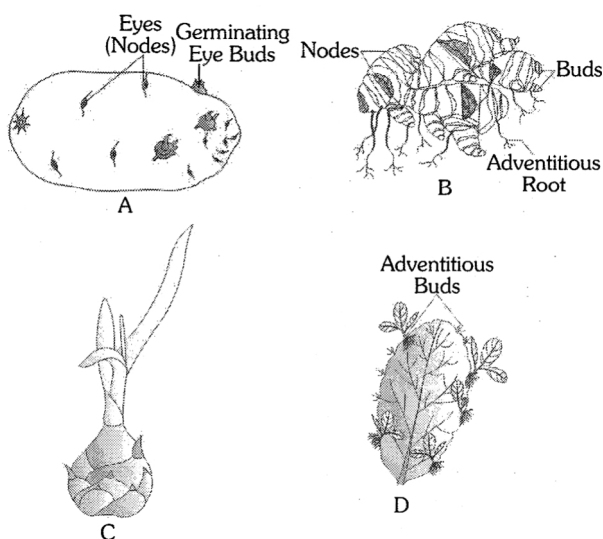
118. Non-albuminous seed is produced in

- (1) Wheat (2) Pea
(3) Maize (4) Castor

119. The mature seeds of plants such as gram and peas, possess no endosperm, because

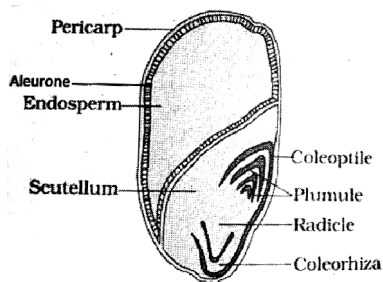
- (1) These plants are not angiosperms
(2) There is no double fertilization in them
(3) Endosperm is not formed in them
(4) Endosperm gets used up by the developing embryo during seed development

120. Study the diagrams given below and select the right options out of (A – D); in which all the 4 items A, B, C and D are correctly identified



- | A | B | C | D |
|------------|---------|--------|-----------|
| (1) Offset | Sucker | Stolon | Leaf buds |
| (2) Offset | Sucker | Stolon | Leaf buds |
| (3) Tuber | Rhizome | Bulb | Leaf buds |
| (4) Tuber | Rhizome | Bulbil | Leaf buds |

121. How many of the labelled structures are diploid in ploidy with respect to the following diagram



- (1) 4 (2) 5
(3) 6 (4) 1

122. Flower is a modified shoot when

- (A) Shoot apical meristem changes to floral meristem.
(B) Internodes do not elongates and the axis gets condensed
(C) The apex produces different kinds of floral appendages laterally at successive nodes.
(D) A shoot tip transforms into a flower, it is always solitary.

- (1) Only A is correct
(2) Only A & B are correct
(3) Only A, B & C are correct
(4) All are correct

123. Assertion : Epiphytes are called space parasites.

Reason : Epiphytic roots possess velamen.

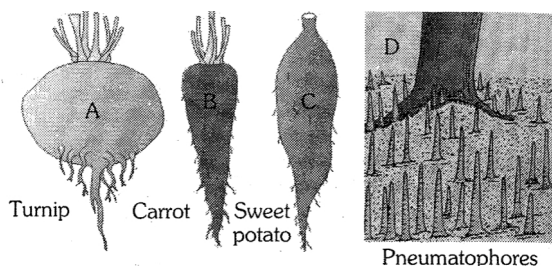
- (1) If both Assertion & Reason are true & the reason is a correct explanation of the assertion.
(2) If both Assertion & Reason are true but reason is not a correct explanation of the arrection.
(3) If Assertion is true but the Reason is false
(4) If both Assertion & Reason are false

124. Assertion : An incomplete flower can be perfect.

Reason : Perfect flowers (incomplete) are neuter.

- (1) If both Assertion & Reason are true & the reason is a correct explanation of the assertion.
(2) If both Assertion & Reason are true but reason is not a correct explanation of the arrection.
(3) If Assertion is true but the Reason is false
(4) If both Assertion & Reason are false

125. See the following diagrams Which of the following is not correct about A, B, C and D



- (1) A, B and C are underground roots but D grows vertically upwards
- (2) Pneumatophore is found in the plants that grow in sandy soil
- (3) Pneumatophores help to get oxygen for respiration
- (4) Tap roots of carrot, turnip and adventitious root of sweet potato, get swollen and store food

126. Which of the following groups of plants are propagated through underground root

- (1) Bryophyllum and kalanchoe
- (2) Ginger, potato, onion and zamikand
- (3) Pistia, chrysanthemum and pineapple
- (4) Sweet potato, asparagus, tapioca and dahlia

127. Vegetative reproduction occurs by bulbil in

- (1) Agave
- (2) Colocasia
- (3) Zingiber
- (4) Vallisneria

128. Which of the following is not a stem modification

- (1) Pitcher of Nepenthes
- (2) Thorns of citrus
- (3) Tendrils of cucumber
- (4) Flattened structures of Opuntia

129. One of single internodal branches are found in

- (1) Asparagus
- (2) Euphorbia
- (3) Lilium
- (4) Casuarina

130. Which of the following is the subaerial stem modification with long internode

- (1) Rhizome
- (2) Offset
- (3) Runner
- (4) Sucker

131. Rachis is present in

- (1) Pinnate compound leaf
- (2) Palmate compound leaf
- (3) Both correct
- (4) Both wrong

132. Onion stores food in

- (1) Underground stem
- (2) Fleshy scales
- (3) Root
- (4) Shoot

133. The whorl of bracts present below the inflorescence of Helianthus (sunflower) is

- (1) Involucre
- (2) Involucel
- (3) Stipule
- (4) Bract

134. Maize grain is

- (1) Seed
- (2) Embryo
- (3) Ovule
- (4) Fruit

135. Which of the following flowers cannot be divided into two equal halves by any vertical plane passing through the centre?

- (1) Cassia and Bean
- (2) Mustard and Datura
- (3) Canna
- (4) Chilli

136. Cockroaches are brown or black bodied animals that are included in classA..... of phylumB..... .

Select correct words for A and B :

A

B

- | | |
|----------------------|------------|
| (1) Insecta | Arthropoda |
| (2) Arthropoda | Insecta |
| (3) Hexapoda | Arthropoda |
| (4) Both (1) and (3) | |

137. Bright yellow, red and green coloured cockroaches have been reported in tropical regions. Their size range from $\frac{1}{4}$ inches to 3 inches. Select correct statements

- (I) Cockroaches have long antenna, legs and flat extension of the upper body wall that conceals head.
- (II) They are nocturnal omnivores that live in damp places throughout the world.
- (III) They have become residents of human homes and and thus are serious pest and vectors of several diseases.
- (IV) *Periplaneta americana* is a common species of cockroach and adults of this species are 34-53 mm long with wings that extend beyond the tip of the abdomen in males.

- (1) Only I, II and III
- (2) Only I and III
- (3) Only II and III
- (4) All of these

138. In a time duration a female cockroach produces 4 ootheca. How many larvae are expected to be produced in this time duration?

- (1) 16 (2) 32
- (3) 8 (4) 64

139. Which of the following structure is found in minimum number in thorax of an adult male cockroach ?

- (1) Segmental ganglia
- (2) Alary muscles
- (3) Spiracles
- (4) Ostia

140. If we want to identify sex of cockroach which part of body should be observed ?

- (1) Head
- (2) Thorax
- (3) Abdomen
- (4) Neck

141. There are two groups of statements given below. Read these statements carefully -

Group - I :

Statement (1) : Frogs can respire in water through skin and through lungs on land.

Statement (2) : Flexible neck is found in cockroach.

Statement (3) : Cloaca is found in only invertebrate animals.

Statement (4) : Tongue of both cockroach and frog is muscular.

Group (II) :

Statement (1) : Head of cockroach shows great mobility in all directions.

Statement (2) : Neck of frog is less flexible than cockroach.

Statement (3) : Limbs of frog are used in walking leaping and burrowing.

How many of the following are incorrect conclusions based upon these statements :

- (A) At least one statement is wrong in both groups.
- (B) There is at least one statement in group (I) which is reason of more than one statements of group (II).
- (C) At least one statement of both groups is related with members of insecta.

- (1) 3 (2) 1
- (3) 2 (4) 0

142. Brain of cockroach is represented by :

- (1) Supra oesophageal ganglion
- (2) Supra pharyngeal ganglion
- (3) Sub oesophageal ganglion
- (4) Sub pharyngeal ganglion

- 143.** Select correct matches of locations and structures of male reproductive system of cockroach :
- (I) Testes — 4th - 6th abdomen of segment
(II) Mushroom glands — 6th - 7th abdomen of segment
(III) Spermatheca — 6th abdomen of segment
- (1) I only (2) I and II only
(3) I, II and III (4) II and III only
- 144.** Hind limbs and fore limbs of frogs can be differentiated
- (1) on the basis of number of digits
(2) on the basis of copulatory pad in case of males
(3) on the basis of size
(4) All of these
- 145.** Select bilobed structures of frog :
- (1) Tongue (2) Gall bladder
(3) Urinary bladder (4) Both (1) and (3)
- 146.** Which of the following fact is true for both frog and human digestive systems :
- (1) Duodenum receives partially digested food from stomach called as chyme.
(2) Duodenum receives bile from gall bladder and pancreatic juice through different openings.
(3) Colon is used mainly for water absorption
(4) More than one options are correct.
- 147.** The vascular system of frog is well developed close type. The blood vascular system includes heart, blood vessels and blood. Which part of frog's heart receives oxygenated blood through conus arteriosus ?
- (1) Right atrium (2) Left atrium
(3) Ventricle (4) None of these
- 148.** Select the paired structures of fore brain of frog :
- (A) Cerebellar hemispheres
(B) Corpora bigemina
(C) Medulla oblongata
- (1) A, B and C (2) A and B
(3) Only B (4) None of these
- 149.** During aestivation which part of heart of frog receives blood with maximum oxygen concentration ?
- (1) Left atrium
(2) Right atrium
(3) Ventricle
(4) No part of heart receives oxygenated blood
- 150.** The abdomen in both male and female cockroaches consist of 10 segments and genital pouch is found in posterior part of abdomen. Consider following openings in genital pouch of cockroaches -
- (A) Anus
(B) Gonopore
(C) Phallic gland opening
(D) Spermathecal opening
(E) Collateral gland opening
- Select appropriate combination of words according to codes to replace X and Y in given statement :
-X..... is the opening which is present at dorsal side of all other openings in case of male andY..... is present ventral to other openings in case of female
- | X | Y | X | Y |
|-------|---|-------|---|
| (1) A | A | (2) A | B |
| (3) C | D | (4) A | D |
- 151.** Which of the following receptors in frog are found in the form of cell aggregations only :
- (1) Photoreceptors
(2) Olfactory receptors
(3) Sound receptors
(4) More than one options are correct
- 152.** The part of skeleton system of frog encloses CNS is :
- (1) Cranium only
(2) Vertebral column only
(3) Skull only
(4) Both (1) and (2)

153. In frog lymph can be differentiated from blood :

- (1) Due to presence of RBC
- (2) Due to presence of WBC
- (3) Both (1) and (2)
- (4) None of these

154. Select correct morphological character of frog :

- (1) Skin smooth and slippery
- (2) Skin of ventral side of body is olive green
- (3) Wall of heart consist of epicardium, myocardium and endocardium
- (4) More than one options are correct

155. Select correct description of RBCs of frog :

- (1) nucleated
- (2) contain haemoglobin
- (3) ovale
- (4) All of these

156. Which of the following statements is correct regarding Cockroach?

- (1) It possesses ventral nerve cord
- (2) Its spiracles help in excretion
- (3) Phallomere is present in female cockroach
- (4) Compound eye is also called as ocellus

157. Select the correct statement from the ones given below with respect to *Periplaneta americana*

- (1) nervous system located dorsally, consists of segmentally arranged ganglia joined by a pair of longitudinal connectives
- (2) Males bear a pair of short thread like anal styles
- (3) There are 16 very long malpighian tubules present at the junction of midgut and hindgut
- (4) Grinding of food is carried out only by the mouth parts

158. If structure "A" of cockroach is found at the junction of "B" and "C" . Which option correctly describes A, B and C :

A	B	C
(1) Foregut	Hepatic caeca	Mid gut
(2) Mid gut	Malpighian tubules	Hind gut
(3) Both (1) and (2)		
(4) None of these		

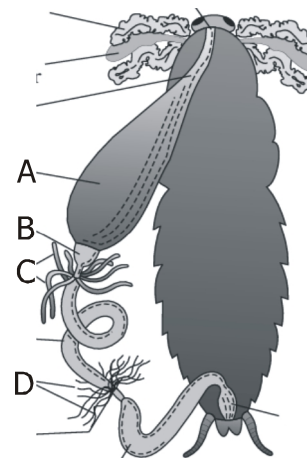
159. Identify the incorrect statement about frog :

- (1) Parathyroid and pineal body are present
- (2) There are ten cranial nerves
- (3) The ventricle opens into the conus arteriosus
- (4) None of these

160. In cockroach, the ootheca is formed by the secretion of

- (1) Phallic gland
- (2) Collateral gland
- (3) Mushroom gland
- (4) Conglobate gland

161. The given figure shows alimentary canal of cockroach. Identify the parts labelled as A to D and select the correct option :



A	B	C	D
(1) Gizzard caecae	Crop tubules	Hepatic	Malpighian
(2) Crop caecae	Gizzard tubules	Hepatic	Malpighian
(3) Crop tubules	Gizzard caecae	Malpighian	Hepatic
(4) Gizzard tubules	Crop caecae	Malpighian	Hepatic

162. Read the given paragraph :

“It is lined by glandular and ciliated cells. It absorbs nitrogenous waste products from haemocoel and convert them into uric acid which is excreted out through the hind gut.”

Which of the following structure of cockroach is referred here?

- (1) Trachea
- (2) Hepatic caecae
- (3) Tergum
- (4) Malpighian tubule

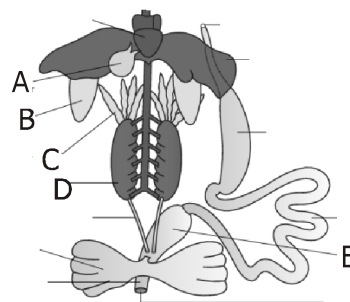
163. Which of the following statements is incorrect about the frog?

- (i) Eyes are bulged out and covered by a nictitating membrane that protect them while in water
 - (ii) On either sides of the eyes a membranous tympanum (ear) receives sound signals
 - (iii) The hind limbs end in four digits and they are larger and muscular than fore limbs that end in five digits
 - (iv) Feet have webbed digits that help in swimming
 - (v) Frogs exhibit sexual dimorphism
- (1) (i) and (v) (2) only (iii)
(3) (ii) and (iii) (4) only (iv)

164. Which of the following statements is correct about the respiration of frog :

- (i) In frog, cutaneous and pulmonary respiration are found
 - (ii) A pair of elongated pink hollow lungs are found in thorax
 - (iii) During aestivation and hibernation, gaseous exchange take place through skin
- (1) (i), (iii) & (iii)
(2) (i) & (iii)
(3) (ii) & (iii)
(4) (i) & (ii)

165. The figure given here shows diagrammatic representation of internal organs of frog. Identify A to E and select the correct option :



A B C D E

- (1) Gall bladder Liver Ovary Testes Rectum
(2) Liver Pancreas Fat bodies Testis Gall bladder
(3) Pancreas Lung Testes Kidney Liver
(4) Gall bladder Lung Fat bodies Kidney Rectum

166. Read the following statements having two blanks A and B :

In cockroach a ring of 6-8 blind tubules called ..(A).. is present at the junction of foregut and midgut while at the junction of midgut and hindgut is present a ring of 100-150 yellow coloured thin filamentous ..(B)..

(A) (B)

- (1) Malpighian tubules Hepatic Caecae
(2) Fat bodies Vasa efferentia
(3) Hepatic caecae Malpighian tubules
(4) Vas deferens Fat bodies

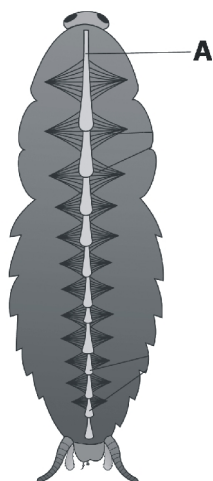
167. The integument of the frog is always kept moist because

- (1) It cannot move with dry skin
- (2) It performs cutaneous respiration
- (3) It cannot catch food with dry skin
- (4) It cannot jump better with moist skin

168. Three chambered heart of frog is not as efficient as four chambered human heart because :

- (1) Oxygenated and deoxygenated bloods mix up
- (2) Ventricle does not pump blood properly
- (3) It does not hold enough blood
- (4) Heart muscles are not strong

- 169.** The cloaca in frog is a common chamber for the urinary tract, reproductive tract and :
- (1) Alimentary canal
 - (2) Notochord
 - (3) Portal system
 - (4) Hepatic portal vessels
- 170.** How many statement(s) are incorrect for circulation of cockroach :
- (I) Heart is present at mid dorsal line of thorax and abdomen.
 - (II) Blood from heart enter in to pericardial sinus through ostia.
 - (III) Blood vessels are absent.
- (1) 3 (2) 1 (3) 2 (4) 0
- 171.** Identify structure 'A' in given figure :



- (1) Crop of an adult cockroach
 - (2) Oesophagus of an adult cockroach
 - (3) Anterior aorta of a female cockroach
 - (4) None of these
- 172.** A frog was being dessected in a training programme. Select the openings in cloaca which can be shown by trainer.
- (1) Rectal opening, urinary bladder opening, urogenital duct opening and pancreatic opening.
 - (2) Rectal opening, urinary bladder opening, urogenital duct opening and oviduct opening.
 - (3) Rectal opening, urinary bladder opening, ureter opening and oviduct opening.
 - (4) More than one options are correct.

- 173.** Alary muscles of coackroach are associated with :
- (1) Respiration (2) Circulation
 - (3) Digestion (4) Locomotion
- 174.** The alimentary canal of frog is short, it indicates :
- (1) It is a carnivores animal
 - (2) It is a herbivores animal
 - (3) It is a terrestrial animal
 - (4) It is an amphibious animal
- 175.** Select correct option in reference of frog :
- (1) Poikilotherm : Body temperature varies with the temp of environment
 - (2) Skin is smooth & slippery due to presence of mucus
 - (3) The frogs never drink water but absorbs it through skin
 - (4) All of these
- 176.** The total number of ganglia in cockroach are
- (1) Three pairs thoracic and three abdominal
 - (2) Three pairs thoracic and six pairs abdominal
 - (3) Three pairs thoracic and eight pairs abdominal
 - (4) Two pairs thoracic and six pairs abdominal
- 177.** The type of circulatory system in *Periplaneta* is
- (1) Lacunar (2) Fenestrated
 - (3) Peritrophic (4) Closed
- 178.** Galea and lacinia are structures present on insect's
- (1) Labrum (2) Labium
 - (3) Antennae (4) Maxillae
- 179.** In cockroach, wings are
- (1) Prothoracic
 - (2) Metathoracic
 - (3) Mesothoracic
 - (4) Both 2 and 3
- 180.** Cockroach has kind of mouth parts
- (1) Lapping and sponging
 - (2) Chewing and biting type
 - (3) Piercing and suking type
 - (4) Chewing and sucking

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